Week 1, Lecture 2
Propagating Fields and Coherent States

Lecturer: Josephine
Email: j.dias@uq.edu.au
Office: Parnell Room 320
Office hours: Thursdays 3-4 pm

Notes 1.2, 1.3, 2.2 and 2.3
Bachor & Ralph, Chapters 2.1, 4.2.2 and 4.4.1
Walls & Milburn, Chapter 2.3

In this lecture…

- Multimode quantization
- Propagating fields
- Coherent states

The University Of Queensland
AUSTRALIA

01

Multimode quantization

From one mode to many modes

A general electromagnetic field contains many modes:

$$\mathbf{A}(\mathbf{r},t) = \sum_j q_j(t)\mathbf{u}_j(\mathbf{r})$$

Each mode j has its own:

- Spatial profile $\mathbf{u}_j(\mathbf{r})$
- Angular frequency ω_j
- Polarisation
- Time-dependent amplitude $q_j(t)$

The modes are orthonormal:

$$\int_V \varepsilon_0 \mathbf{u}_j(\mathbf{r}) \cdot \mathbf{u}_k(\mathbf{r})\, \mathrm{d}^3\mathbf{r} = \delta_{jk}$$

Each mode is an independent oscillator

The classical field Hamiltonian becomes

$$\boxed{H_{\mathrm{EM}} = \sum_j \left(\frac{1}{2} p_j^2 + \frac{1}{2} \omega_j^2 q_j^2 \right)}$$

Canonical quantization gives

$$q_j \longrightarrow \hat{q}_j, \qquad p_j \longrightarrow \hat{p}_j$$

The operators satisfy

$$\boxed{[\hat{q}_j, \hat{p}_k] = i\hbar\delta_{jk}}$$

Creation and annihilation operators for each mode

Each mode has its own creation and annihilation operators:

$$\hat{a}_j = \sqrt{\frac{\omega_j}{2\hbar}}\,\hat{q}_j + \frac{i}{\sqrt{2\hbar\omega_j}}\,\hat{p}_j$$

$$\hat{a}_j^\dagger = \sqrt{\frac{\omega_j}{2\hbar}}\,\hat{q}_j - \frac{i}{\sqrt{2\hbar\omega_j}}\,\hat{p}_j$$

They satisfy

$$\boxed{[\hat{a}_j, \hat{a}_k^\dagger] = \delta_{jk}}$$

and

$$[\hat{a}_j, \hat{a}_k] = 0, \qquad [\hat{a}_j^\dagger, \hat{a}_k^\dagger] = 0$$

The multimode Hamiltonian

Define the number operator for mode j:

$$\hat{n}_j = \hat{a}_j^\dagger \hat{a}_j$$

The multimode Hamiltonian is

$$\hat{H}_{\mathrm{EM}} = \sum_j \hbar\omega_j \left(\hat{n}_j + \frac{1}{2}\right)$$

The quantized vector potential is

$$\hat{\mathbf{A}}(\mathbf{r}) = \sum_j \sqrt{\frac{\hbar}{2\omega_j}} \left(\hat{a}_j + \hat{a}_j^\dagger\right) \mathbf{u}_j(\mathbf{r})$$

Multimode number states

A multimode number state is written as

$$|n_1, n_2, n_3, \ldots\rangle = |n_1\rangle \otimes |n_2\rangle \otimes |n_3\rangle \otimes \cdots$$

Each number operator acts on its corresponding mode:

$$\hat{n}_j |n_1, \ldots, n_j, \ldots\rangle = n_j |n_1, \ldots, n_j, \ldots\rangle$$

The multimode vacuum is

$$|0, 0, 0, \ldots\rangle$$

Examples of multimode states

For two modes:

$$|0,0\rangle = \text{vacuum}$$
$$|1,0\rangle = \text{one photon in mode 1}$$
$$|0,1\rangle = \text{one photon in mode 2}$$
$$|2,0\rangle = \text{two photons in mode 1}$$
$$|1,1\rangle = \text{one photon in each mode}$$

The total photon-number operator is

$$\boxed{\hat{N} = \sum_j \hat{n}_j}$$

For a multimode number state,

$$\hat{N}|n_1, n_2, \ldots\rangle = \left(\sum_j n_j\right)|n_1, n_2, \ldots\rangle$$

Some points on the multimode field

For a multimode electromagnetic field,

$$\hat{H} = \sum_j \hbar\omega_j \left(\hat{n}_j + \frac{1}{2}\right)$$

The multimode vacuum is $|\mathrm{vac}\rangle = |0, 0, 0, \ldots\rangle$

Its energy is

$$E_{\mathrm{vac}} = \frac{1}{2}\sum_j \hbar\omega_j$$

Because there are infinitely many modes,

$$E_{\mathrm{vac}} \to \infty$$

In quantum optics, we often normal order the Hamiltonian:

$$:\hat{H}: = \sum_j \hbar\omega_j \hat{a}_j^\dagger \hat{a}_j$$

so that $\langle \mathrm{vac}| :\hat{H}: |\mathrm{vac}\rangle = 0$

The University Of Queensland
AUSTRALIA

02

Propagating fields

Plane-wave basis

For a standing-wave mode,

$$\hat{\mathbf{E}}(\mathbf{r},t) = \sum_m \sqrt{\frac{\hbar\omega_m}{\epsilon_0 V}}\,\mathbf{e}_m \left[\hat{a}_m(0)e^{-i\omega_m t} + \hat{a}_m^\dagger(0)e^{i\omega_m t}\right]\sin(k_m z)$$

For freely propagating light, use travelling-wave modes:

$$\hat{\mathbf{E}}(\mathbf{r},t) = \sum_m \sqrt{\frac{\hbar\omega_m}{2\epsilon_0}} \left[\mathbf{u}_m(\mathbf{r})\hat{a}_m(0)e^{-i\omega_m t} + \mathbf{u}_m^*(\mathbf{r})\hat{a}_m^\dagger(0)e^{i\omega_m t}\right]$$

with $\mathbf{u}_m(\mathbf{r}) = \dfrac{\mathbf{e}_m}{\sqrt{V}} e^{i\mathbf{k}_m\cdot\mathbf{r}}$

For a periodic quantization volume $V = L^3$, $\mathbf{k}_m = \dfrac{2\pi}{L}\left(m_x\hat{\mathbf{x}} + m_y\hat{\mathbf{y}} + m_z\hat{\mathbf{z}}\right), \qquad m_x, m_y, m_z \in \mathbb{Z}$

Standing and travelling waves are different mode bases for describing the same electromagnetic field.

A single travelling mode

Consider one mode with:

- Angular frequency ω
- Wavevector $\mathbf{k} = k\hat{\mathbf{z}}$
- A single transverse polarization $\mathbf{e}$
- Phase $\phi = kz - \omega t$

For propagation in vacuum,

$$\omega = ck$$

The electric-field operator is

$$\hat{\mathbf{E}}(z,t) = \mathbf{e}E_0\left(\hat{a}e^{i\phi} + \hat{a}^\dagger e^{-i\phi}\right)$$

where $E_0 = \sqrt{\dfrac{\hbar\omega}{2\epsilon_0 V}}$

The mode operators satisfy $[\hat{a}, \hat{a}^\dagger] = 1$

Electric field and vector potential

In the Coulomb gauge, for a source-free transverse field,

$$\mathbf{E}(\mathbf{r},t) = -\frac{\partial \mathbf{A}(\mathbf{r},t)}{\partial t}$$

Therefore, for our single mode,

$$\hat{\mathbf{A}}(z,t) = -\frac{iE_0}{\omega}\mathbf{e}\left(\hat{a}e^{i\phi} - \hat{a}^\dagger e^{-i\phi}\right)$$

And we can see that

$$[\hat{E},\hat{A}] = \frac{2iE_0^2}{\omega} = \frac{i\hbar}{\epsilon_0 V} \neq 0$$

Gaussian mode

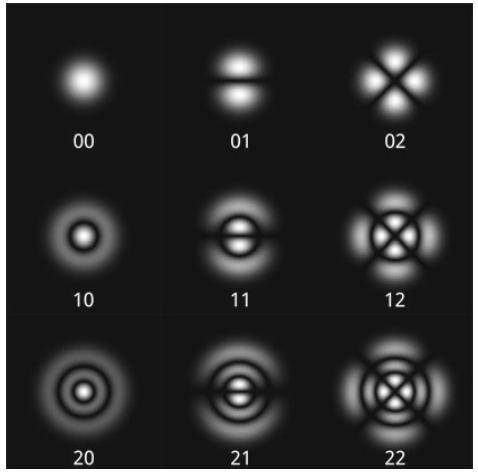
A plane wave extends infinitely in the transverse direction and is therefore an idealization.

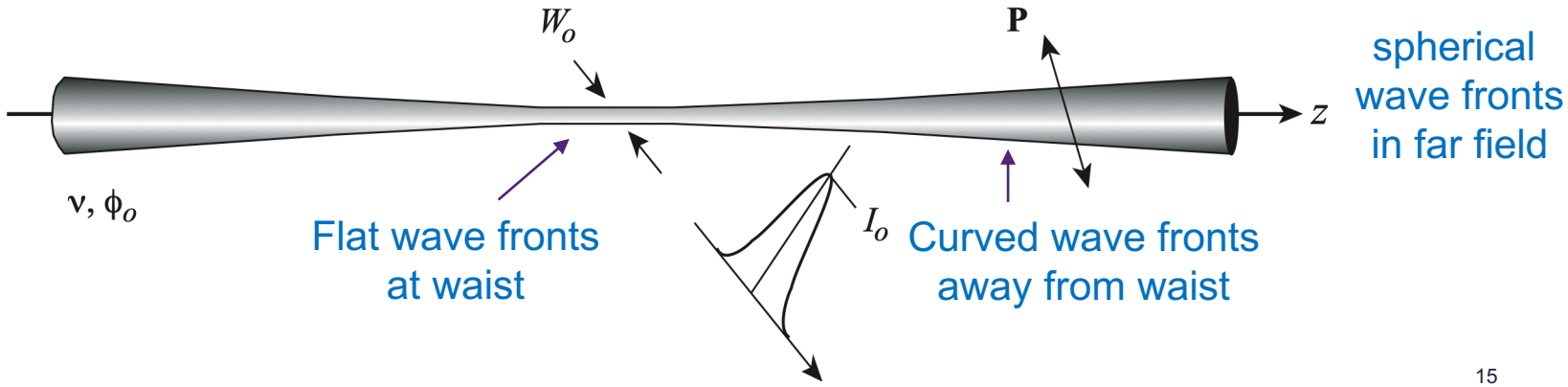
Real laser beams have a finite transverse width.

The simplest model of a finite-width laser beam is a Gaussian mode.

The beam is narrowest at the beam waist.

The wavefront is approximately flat near the waist.





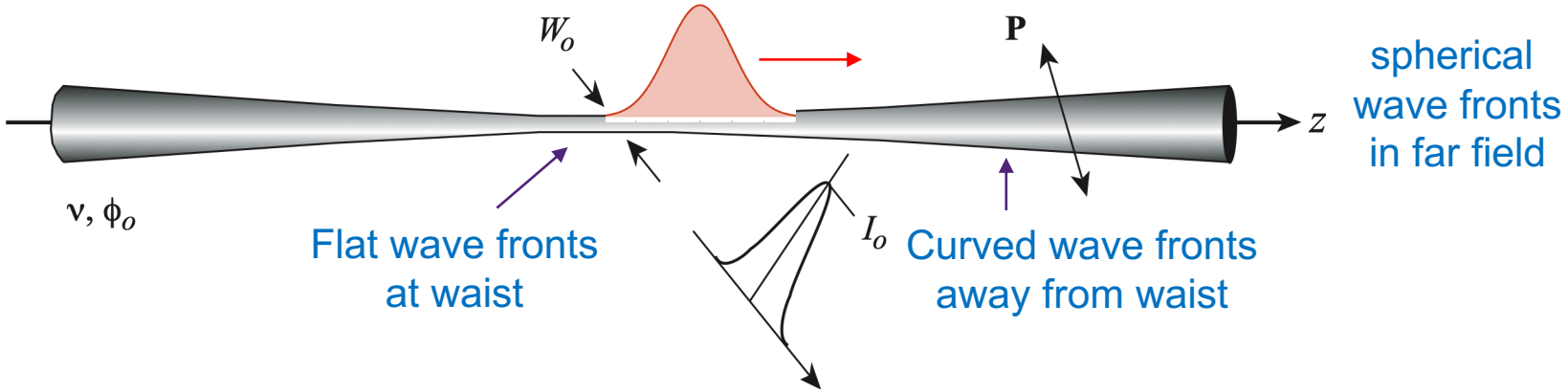
Localized Pulse

Consider a single frequency and single polarization mode plane wave propagating in the *z*-direction

$$\hat{E} = E_o(\hat{a}e^{i(kz-\omega t)} + \hat{a}^\dagger e^{-i(kz-\omega t)})$$

But single frequency implies that the pulse length is infinite

Consider instead a Localized pulse



To construct a pulse we need multiple frequencies – how do we model this?

Continuum of frequency modes

In free space, frequency modes form a continuum.
A pulse travelling in the z-direction can be written as

$$\hat{\mathbf{E}}(z,t) = E_0 \mathbf{e}_m \int d\omega \left[F(\omega)\hat{a}_\omega e^{i[k(\omega)z-\omega t]} + F^*(\omega)\hat{a}_\omega^\dagger e^{-i[k(\omega)z-\omega t]} \right]$$

$F(\omega)$ is the spectral mode function.
$\hat{a}_\omega$ annihilates a photon at frequency ω.

Normalization and commutation relation

The spectral mode function is normalized: $\int d\omega \, |F(\omega)|^2 = 1$

The continuum operators obey $\left[\hat{a}_\omega, \hat{a}_{\omega'}^\dagger\right] = \delta(\omega - \omega')$

The Dirac delta is the continuum equivalent of the Kronecker delta.

Different frequency modes are independent.

Pulse-mode operator

Define an annihilation operator for the complete pulse:

$$\hat{a}_F(z,t) = \int d\omega\, F(\omega)\hat{a}_\omega e^{i[k(\omega)z-\omega t]}$$

It obeys the usual single-mode commutation relation:

$$\left[\hat{a}_F, \hat{a}_F^\dagger\right] = \int d\omega\, |F(\omega)|^2 = 1$$

The field can therefore be written compactly as

$$\hat{\mathbf{E}}(z,t) = E_0 \mathbf{e}_m \left(\hat{a}_F + \hat{a}_F^\dagger\right)$$

The University Of Queensland
AUSTRALIA

03

Coherent states

Why introduce coherent states?

Number states describe states with an exact photon number:

$$\hat{n}|n\rangle = n|n\rangle$$

However, a number state has no definite optical phase and its average electric field is zero:

$$\langle n|\hat{E}(t)|n\rangle = 0$$

Coherent states provide the closest quantum description of a classical electromagnetic wave.

An ideal, stable laser field is approximately described by a coherent state.

In a coherent state,

$$\langle\alpha|\hat{E}(t)|\alpha\rangle = \mathcal{E}_0\left(\alpha e^{-i\omega t} + \alpha^* e^{i\omega t}\right)$$

Equivalently,

$$\langle\hat{E}(t)\rangle = 2\mathcal{E}_0|\alpha|\cos(\omega t - \arg\alpha)$$

Definition of a coherent state

A coherent state is an eigenstate of the annihilation operator:

$$\boxed{\hat{a}|\alpha\rangle = \alpha|\alpha\rangle}$$

where $\alpha \in \mathbb{C}$

We can write $\alpha = |\alpha|e^{i\theta}$

- |α| determines the field amplitude and mean photon number.
- $\theta = \arg\alpha$ determines the optical phase.

Is it also an eigenstate of $\hat{a}^\dagger$?

A coherent state is not an eigenstate of the creation operator:

$$\boxed{\hat{a}^\dagger|\alpha\rangle \neq \alpha^*|\alpha\rangle}$$

However, taking the adjoint of

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$$

gives

$$\boxed{\langle\alpha|\hat{a}^\dagger = \alpha^*\langle\alpha|}$$

Therefore:

- $|\alpha\rangle$is a right eigenstate of $\hat{a}$.
- $\langle\alpha|$is a left eigenstate of $\hat{a}^\dagger$.
- $|\alpha\rangle$is not a right eigenstate of $\hat{a}^\dagger$.

Coherent states in the number basis

Expand the coherent state in the number-state basis:

$$|\alpha\rangle = \sum_{n=0}^{\infty} c_n |n\rangle$$

Apply the annihilation operator:

$$\hat{a}|\alpha\rangle = \sum_{n=1}^{\infty} c_n \sqrt{n}\,|n-1\rangle$$

Relabelling $m = n - 1$,

$$\hat{a}|\alpha\rangle = \sum_{m=0}^{\infty} c_{m+1} \sqrt{m+1}\,|m\rangle$$

Using $\hat{a} \mid \alpha\rangle = \alpha \mid \alpha\rangle$,

$$c_{m+1}\sqrt{m+1} = \alpha c_m$$

Therefore,

$$c_n = \frac{\alpha^n}{\sqrt{n!}} c_0$$

Normalised coherent state

Normalisation requires

$$\langle\alpha|\alpha\rangle = 1$$

Therefore,

$$1 = |c_0|^2 \sum_{n=0}^{\infty} \frac{|\alpha|^{2n}}{n!} = |c_0|^2 e^{|\alpha|^2}$$

Choosing c_0 to be real and positive gives

$$c_0 = e^{-|\alpha|^2/2}$$

The normalised coherent state is

$$\boxed{|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle}$$

For example,

$$|\alpha\rangle = e^{-|\alpha|^2/2} \left(|0\rangle + \alpha|1\rangle + \frac{\alpha^2}{\sqrt{2!}}|2\rangle + \frac{\alpha^3}{\sqrt{3!}}|3\rangle + \cdots \right)$$

The displacement operator

The displacement operator is defined by

$$\boxed{\hat{D}(\alpha) = \exp\left(\alpha\hat{a}^\dagger - \alpha^*\hat{a}\right)}$$

The exponent is anti-Hermitian:

$$\left(\alpha\hat{a}^\dagger - \alpha^*\hat{a}\right)^\dagger = -\left(\alpha\hat{a}^\dagger - \alpha^*\hat{a}\right)$$

Therefore, $\hat{D}(\alpha)$ is unitary:

$$\boxed{\hat{D}^\dagger(\alpha) = \hat{D}^{-1}(\alpha) = \hat{D}(-\alpha)}$$

A coherent state is obtained by applying the displacement operator to the vacuum:

$$\boxed{|\alpha\rangle = \hat{D}(\alpha)|0\rangle}$$

Recovering the coherent-state expansion

Using the Baker–Campbell–Hausdorff relation,

$$\boxed{\hat{D}(\alpha) = e^{-|\alpha|^2/2} e^{\alpha \hat{a}^\dagger} e^{-\alpha^* \hat{a}}}$$

Acting on the vacuum gives

$$\hat{D}(\alpha)|0\rangle = e^{-|\alpha|^2/2} e^{\alpha \hat{a}^\dagger}|0\rangle$$

because

$$e^{-\alpha^* \hat{a}}|0\rangle = |0\rangle$$

Expanding the remaining exponential,

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{n!} (\hat{a}^\dagger)^n |0\rangle$$

Using

$$(\hat{a}^\dagger)^n |0\rangle = \sqrt{n!}|n\rangle$$

recovers

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle$$

Properties of displacement operators

The displacement operator transforms the annihilation and creation operators as

$$\boxed{\hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha) = \hat{a} + \alpha}$$

and

$$\boxed{\hat{D}^\dagger(\alpha)\hat{a}^\dagger\hat{D}(\alpha) = \hat{a}^\dagger + \alpha^*}$$

Two displacements combine according to

$$\boxed{\hat{D}(\alpha)\hat{D}(\beta) = e^{(\alpha\beta^* - \alpha^*\beta)/2}\hat{D}(\alpha + \beta)}$$

Therefore,

$$\hat{D}(\alpha)|\beta\rangle = e^{(\alpha\beta^* - \alpha^*\beta)/2}|\alpha + \beta\rangle$$

Coherent states are not orthogonal

The overlap between two coherent states is

$$\boxed{\langle\alpha|\beta\rangle = \exp\left[-\frac{|\alpha|^2}{2} - \frac{|\beta|^2}{2} + \alpha^*\beta\right]}$$

Therefore,

$$|\langle\alpha|\beta\rangle|^2 = e^{-|\alpha-\beta|^2}$$

Consequences:

- Different coherent states are not perfectly distinguishable.
- Their overlap decreases as their phase-space separation increases.
- Coherent states form an overcomplete basis:

$$\boxed{\frac{1}{\pi}\int_{\mathbb{C}} |\alpha\rangle\langle\alpha|\, d^2\alpha = \hat{I}}$$

Summary

The electromagnetic field can be described as a collection of independent modes.

Each mode is quantized as a harmonic oscillator:

$$\hat{H} = \sum_j \hbar\omega_j \left(\hat{n}_j + \frac{1}{2}\right)$$

Coherent states are eigenstates of the annihilation operator:

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$$

Coherent states have a nonzero mean electric field and most closely resemble classical light.